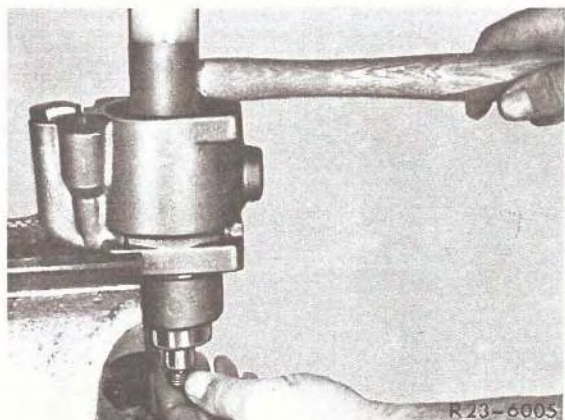


ZF 7673

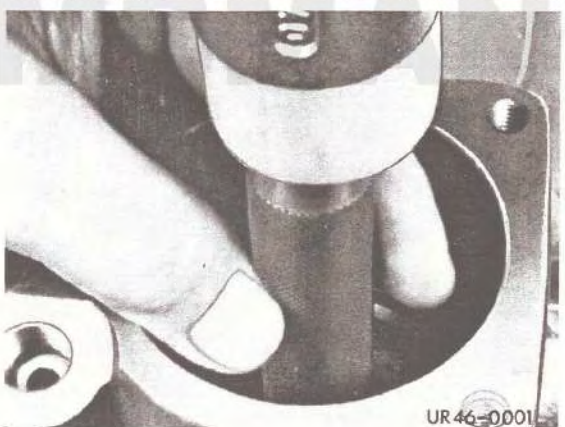


6 Knock drive shaft out of body.



7 Using a soft drift, force face plate at drive end out of body.

8 Remove, disassemble and check flow limiting valve with reference to 3.1/1.



9 Knock needle bearing and shaft seal jointly out of body, using special tool No. 1.

10 Take O-rings out of radial grooves in body and remove dowel pin. Take O-ring off face plate at drive end.

11 If ball bearing is damaged, remove snap ring and then force ball bearing off drive shaft.

12 Check pump components, referring to 4.1/1.

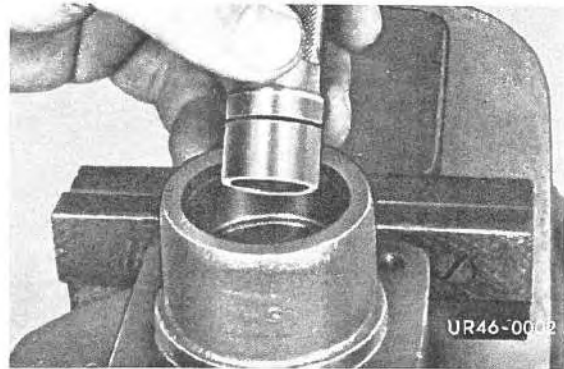
Assembly

Note: Prior to assembling, clean all parts, apply thin film of ATF and exchange all sealing rings and O-rings.

Preassembly of pump body

1 Using special tool No. 1, press needle bearing into body until gap of 37,0 to 37,2 mm is left between end of needle bearing and end of body.

Note: Only pumps with V-belt drive incorporate needle bearing.



2 Press shaft seal into body, using special tool No. 2. Fit seal with high-temperature bearing grease between lips.

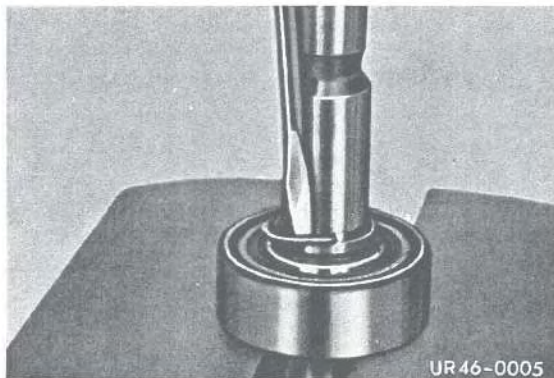
3 Screw socket-head plug into pump body and tighten.



4 Position O-ring in bottom most radial groove inside the body.

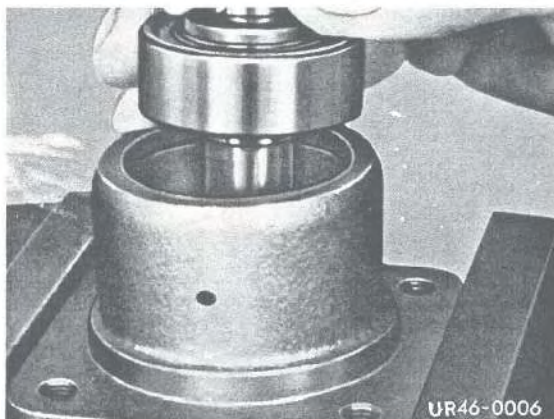
5 Fit dowel pin.



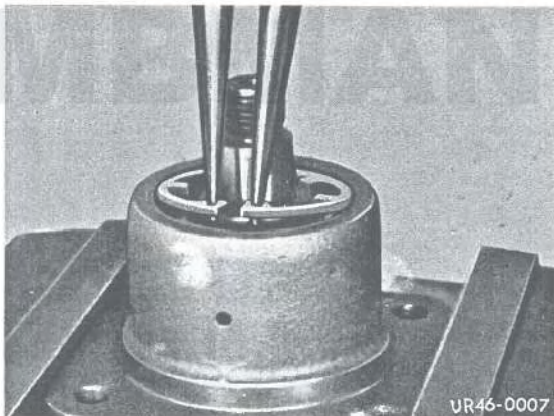


Assembly and installation of drive shaft

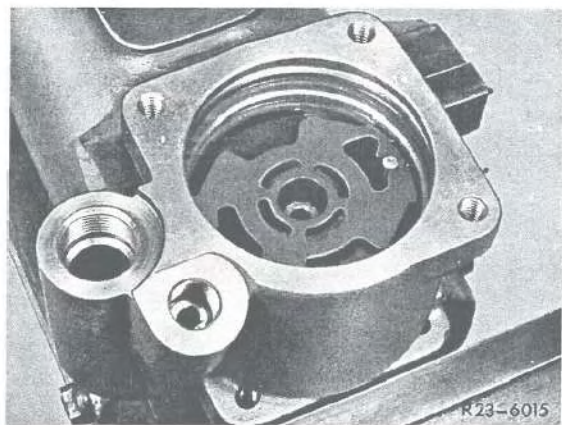
6 Press ball bearing onto drive shaft. Fit snap ring and Woodruff key.



7 Press preassembled drive shaft into body until ball bearing contacts collar.



8 Fit snap ring of deep-groove ball bearing.



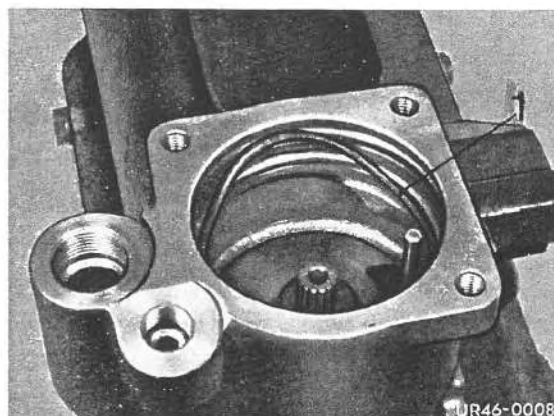
Installation of face plate at drive end

9 Introduce O-ring into radial groove in face plate neck.

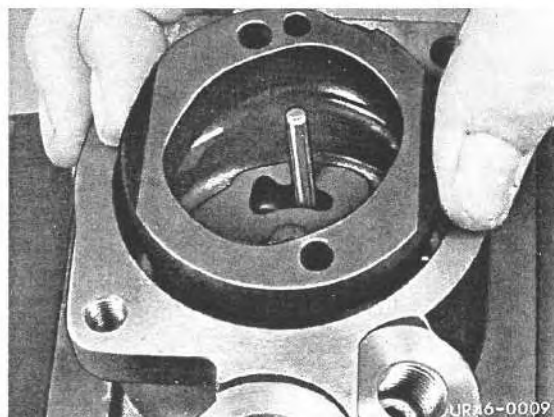
10 Slip face plate onto pin, pressing at neck into bore.

Installation of cam ring and rotor

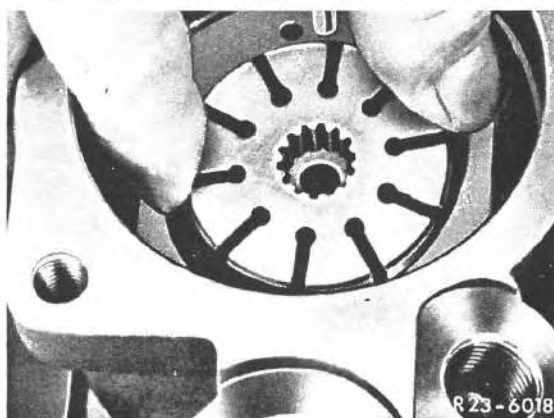
11 Fit O—ring in radial body groove.



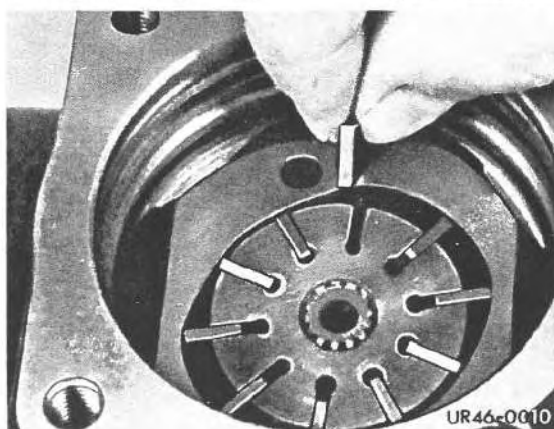
12 Insert cam ring. Smaller bore encompasses straight pin whereas 2 larger bores are situated above face plate passages. The half arrow in the casting points in pump drive direction.



13 Slip rotor onto drive shaft, with smooth bore first.



14 Slip ten vanes into rotor slots, with round outer faces pointing toward cam ring.

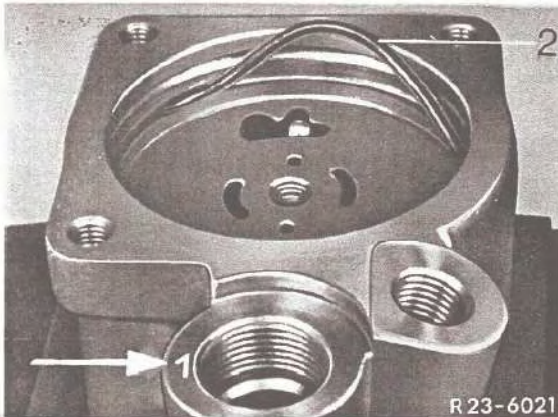


ZF 7673



Installation of face plate and companion cover

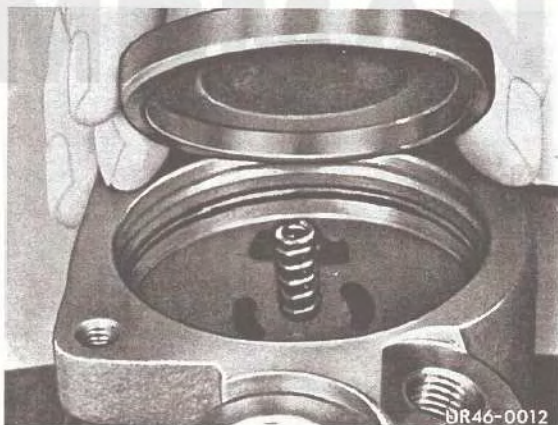
15 Press face plate onto cam ring. Pin fits in one of the two smaller bores.



16 Insert O-ring.

1 O-ring

2 Tolerance group



17 Put compression ring in face plate bore. Position cover with recess pointing inward, and press into body until hooked snap ring can be prised in. Fit snap ring.

18 Install flow limiting valve, referring to 3.1/1.

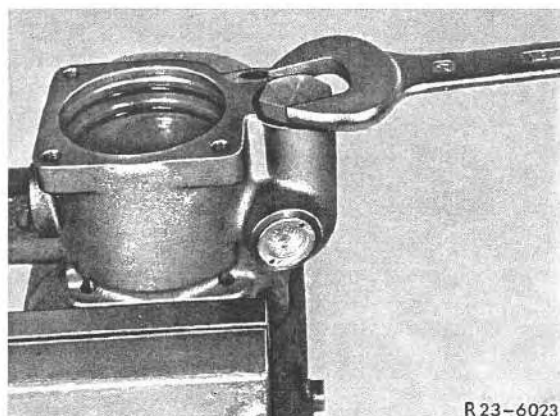
Removal

- 1 Unscrew plug
- 2 Remove compression spring and valve piston.

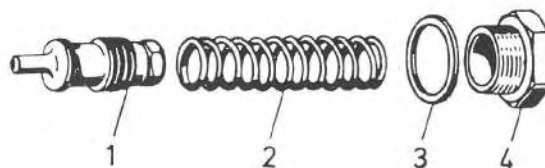
- 1 Valve
- 2 Compression spring
- 3 Washer
- 4 Screw plug

Testing

- 3 Check flow limiting valve for scoring and wear. Flow control piston must not stick in bore. Make sure tolerance group number on body (at end of valve bore) tallies with that on piston.



R 23-6023



Z 46-6032

Installation

- 4 Clean control valve and screw, blowing out with compressed air. Put back compression spring, guide pin and ball. After placing washers in position, turn valve seating screw into control piston and tighten.

- 5 Insert assembled valve piston into bore, with thin stem first. Make sure tolerance groups tally.



R 23-6024

- 6 Slip compression spring onto valve piston and screw plug with sealing ring into tapped hole.



R 23-6025

Checking

- 1 Check all parts for signs of wear.
- 2 Check drive shaft for scoring caused by shaft seal and needle bearing.
- 3 Check needle bearing in body for wear, exchanging if necessary (only pumps featuring V-belt drive).
- 4 Check face plates at drive and cover ends for wear and scoring.
- 5 Check rotor and cam ring for wear. These two parts are always to be exchanged together with pump vanes.
- 6 Check pump vanes for scoring and wear, especially on round running surface.
- 7 Blow compressed air through all bores in body and connections.

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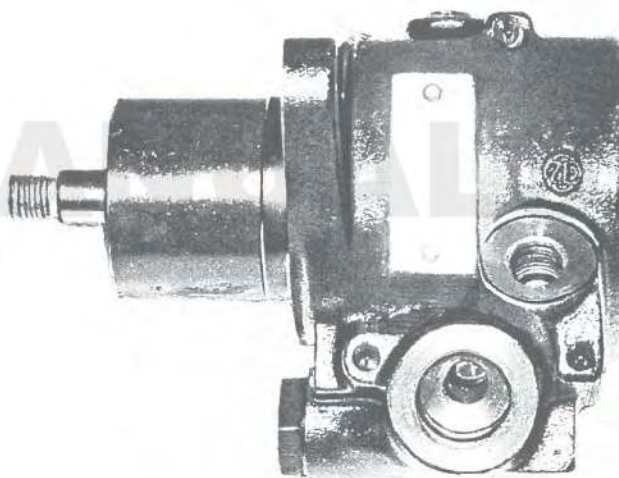
Chapter	Page
1 General	
Installation survey	1.1/1
General view	1.1/1
Technical data	1.2/1
Oil grades	1.2/1
Functional diagram	1.3/1
Faults and remedies	1.4/1
Exploded view	1.5/1

Installation survey

Chassis Model	Sales designation	Steering Model	Sales designation	Power steering pump		Installation
				Sales designation	Type	
435.110	U 1700	765.601	LS 3 B	ZF 7672 ¹⁾	ZF vane pump with built-in current and pressure limiting valve	Standard
.111	U 1700 L					
.113	U 1700 L/38					
.115	U 1300 L					
.117	U 1300 L/37					

1) Daimler-Benz part No. 001 466 03 01

General view



UR46-0269

ZF vane pump 7672

Technical data

Controlled delivery	Q	L/min	12	
	at engine speed	n	1/min	780 to 870 650 to 720 ¹⁾
	and at pump speed	n	1/min	900 to 1000
	at pump pressure	p	bar	130 + 5
	at oil temperature	t	°C	50 to 60
Max. pressure	p max	bar	130 + 5	
Speed	n max at engine speed 2800/min	1/min	3200	3800 ¹⁾
Pump rotation			clockwise	
Pump drive	Pulley dia.	d	mm	161.6 138.5 ¹⁾
	V-belt		mm	12.5 x 925 ± 5 ¹⁾
			mm	12.5 x 955

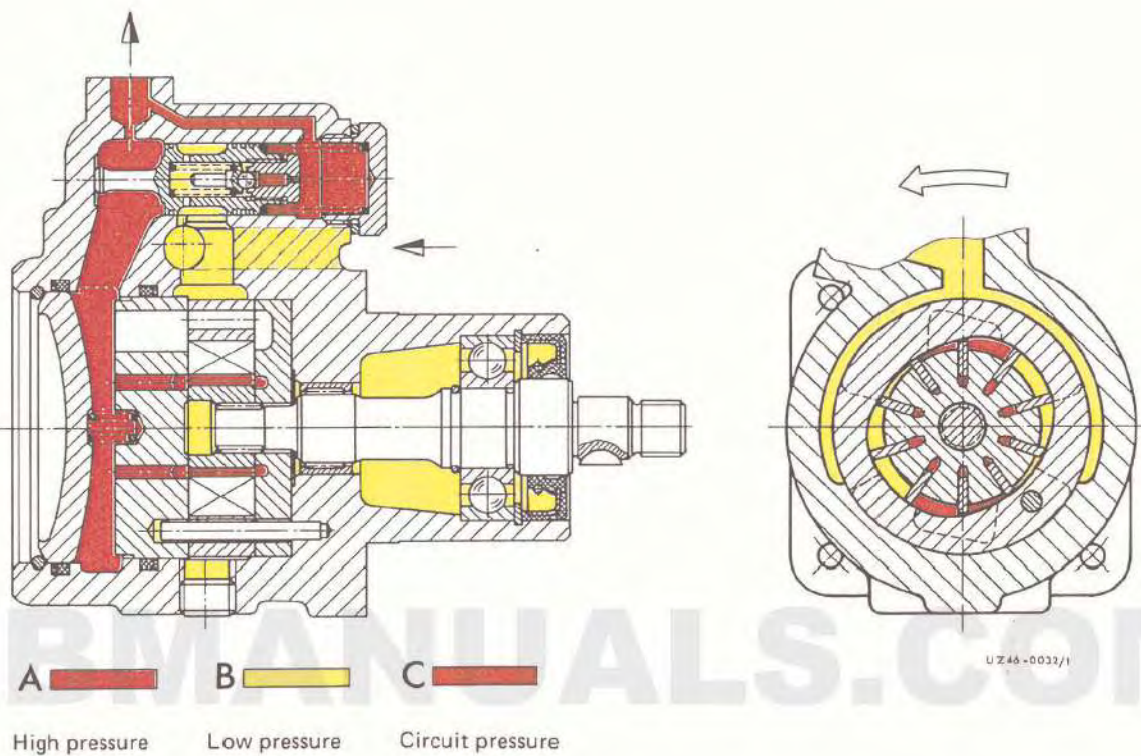
¹⁾ Optional for special-purpose vehicles

Oil grades

Outdoor temperatures		Viscosity	
Zone	Remarks	Summer	Winter
Moderate		SAE 10 W ATF ^{*)}	SAE 10 W ATF ^{*)}
Cold	Essential to change oil in summer and winter (does not apply to ATF)	SAE 10 W ATF ^{*)}	SAE 5 W – 20 H/30 ATF ^{*)}
Hot	Temperature in oil tank must not exceed 65°C	SAE 30	SAE 30

^{*)} ATF must not be blended with engine oil

Functional diagram



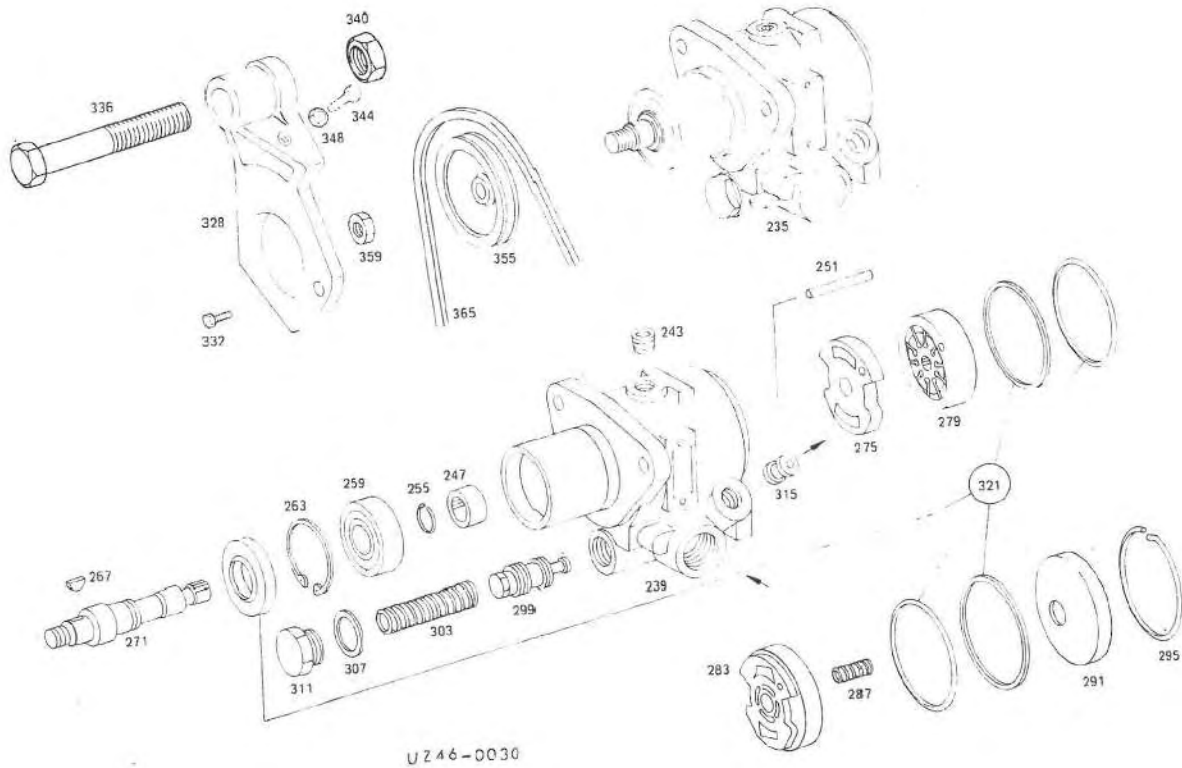
ZF vane pump 7672

Faults and Remedies

Fault	Cause	Remedy
Pump noisy.	Inadequate oil quantity in system (air being sucked in)	While engine is running, raise oil level to upper mark on dipstick and then bleed system.
	Pump sucks air in.	Tighten suction line connections. Renew sealing ring if necessary.
Pressure fails.	Valve piston sticking.	Clean valve piston and check for freedom of movement in valve bore. Fit new valve if necessary. Note tolerance stage.
	Face plate at cover end does not contact cam ring. (Face plate sticking or foreign matter in assembly).	Clean face plate and check for scoring.
	O-rings damaged or forgotten.	Insert new O-rings.
Maximum pressure setting not reached.	V-belt slack or defective.	Tighten or renew V-belt. Thumb test.
	Flow limiting valve fails to close.	Remove and clean flow limiting valve. Throttle ports must not be clogged. Use new valve if necessary. Note tolerance group.
	Pump vanes or cam ring worn.	Fit new set of pump vanes together with new cam ring and rotor. Contact surfaces of face plates at cover and drive ends are also to be checked for wear and damage. Exchange if necessary.
	Rotor shows too much end play.	Renew rotor, cam ring and vanes jointly.
	Foreign matter at face plate.	Clean face plate and check for scoring.
	Flow limiting valve maladjusted.	Exchange flow limiting valve assy. Note tolerance stages.
	Throttle port clogged or pressed in (maximum pressure reached only at very low delivery).	Clean throttle insert.
	O-ring damaged.	Exchange O-ring.

Fault	Cause	Remedy
Specified delivery not attained.	V-belt slack or damaged.	Tighten or replace V-belt. Thumb test.
	Valve piston sticking or spring is too weak.	Clean piston of flow limiting valve and check for freedom of movement in valve bore. If necessary fit new spring or new valve. Note tolerance group.
	Flow limiting valve fails to close.	Remove piston of flow limiting valve. Release valve screw and clean all parts. Fit new valve if necessary. Note tolerance group.
	Throttle port clogged..	Clean throttle insert.
	Pump vanes or cam ring worn.	Fit new set of pump vanes together with new cam ring and rotor. Contact surfaces of face plates at cover and drive ends are also to be checked for wear and damage. Exchange if necessary.
	Rotor shows too much end play.	Renew rotor, cam ring and vanes jointly.
Pump delivers too much oil.		
Delivery keeps increasing with increasing speed and rises far beyond specified max. control flow.	Valve piston sticking.	Find obstruction and rectify. Foreign matter may be responsible. Replace valve if necessary. Note tolerance group.
Delivery increases with increasing speed beyond specified max. control flow but, at a given speed, attains a max. value which will no longer increase even at higher speeds.	Compression spring too hard.	Replace compression spring.
Max. pressure exceeded.	Bore in flow limiting valve clogged.	Clean parts; fit new valve if necessary. Check max. pressure.
	A washer was forgotten after valve cleaning.	Fit washer. Check max. pressure. Fit new valve if necessary.

Exploded view



Power steering pump

235	Power steering pump	295	Snap ring
239	Body	299	Piston
243	Screw plug	303	Spring
247	Needle bearing	307	Sealing ring
251	Pin	311	Screw plug
255	Snap ring	328	Holder
259	Ball bearing	332	Bolt
263	Snap ring	336	Bolt
267	Woodruff key	340	Nut
271	Drive shaft	344	Bolt
275	Cover	348	Nut
279	Gear	355	Pulley
283	Cover	359	Nut
287	Spring	365	V-belt
291	Cover		

Survey

Version

Electrical System

54.7

Electrical System 435.115 with SA 35 769/11

54.12

Electrical System 435.115 with box-type body

54.13

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Hydrostat	1.7/1
Rotating Beacon	1.8/1
Auxiliary Headlamp	1.9/1
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Installation Survey of Circuit Diagrams

Chassis Model	Sales designation	Electrical equipment Version	Installation chassis end No.	Series/SA	Remarks Group Page
435.110 .111 .113	U 1700 U 1700 L U 1700 L/38	Electrical circuit diagram with external regulator	to 001 067	Series	1.5/2
.115 .117	U 1300 L U 1300 L/37	Electrical circuit diagram with integrated regulator 12 V system 24 V system	from 001 068	Series SA 35 769/5	1.5/4
435.111	U 1700 L	Electrical circuit diagram Masked circuit and 24 V system with external regulator	to 001 517	SA 35 769/10	1.5/6
.115	U 1300 L	Electrical circuit diagram Masked circuit and 24 V system with integrated regulator	from 001 518	SA 35 769/10	1.5/8
435.113	U 1700 L/38	Electrical circuit diagram 24 V system with masked circuit		SA 35 769/14	1.5/10
435.113	U 1700 L/38	Electrical circuit diagram 24 V system		SA 35 769/16	1.5/12
.115	U 1300 L			SA 35 769/15	
435.113	U 1700 L	Electrical circuit diagram		SA 35 769/17	1.5/14
435.110 .111 .115	U 1700 U 1700 L U 1300 L	Electrical circuit diagram Windscreen heated 12 V system		SA 35 962/1	1.6/1
.117	U 1300 L/37	Electrical circuit diagram Windscreen heated 24 V system		SA 35 962/2	1.6/2

54.7 General

Installation Survey Circuit Diagrams

Chassis Model	Sales designation	Electrical equipment Version	Installation Chassis end No.	Series/SA	Remarks Group Page
435.111 .115	U 1700 L U 1300 L	Electrical circuit diagram Hydrostat		SA 36 017	1.7/1
435.110 .111 .113 .115 .117	U 1700 U 1700 L U 1700 L/38 U 1300 L U 1300 L/37	Electrical circuit diagram Rotating beacon		SA 35 941	1.8/1
435.110 .111 .113 .115 .117	U 1700 U 1700 L U 1700 L/38 U 1300 L U 1300 L/37	Circuit diagram auxiliary headlamp		SA 53 496	1.9/1
435.110 .111 .115 .117	U 1700 U 1700 L U 1300 L U 1300 L/37	Circuit diagram Air conditioning system 12 V system Air conditioning system 24 V system		SA 35 914 SA 53 970 SA 35 914 SA 53 970	1.10/1 1.10/2

Technical Data**Battery**

Battery		Series	SA 35 769/5	SA 35 979
Voltage volt		12	24	24
Capacity Ah		120	2 x 55	2 x 125
Low-temperature test current A		420	255	625
Electrolyte level above upper edge of plate mm		10 – 12		
Charging current A	at initial charge	max. 5%		
	normal during recharging	* max. 10% of battery capacity		
	at fast charge	to 75%		
Maximum temperature		40 ° C		
Maximum temperature	Tropics	50 ° C		
Freezing point	well charged	–70 ° C (for tropics– 40 ° C)		
	half charged	–25 ° C (for tropics– 13 ° C)		
	discharged	–8 ° C (for tropics– 6 ° C)		

* at commencement of gassing falling to 5%

Electrolyte density (normal)

The operating condition of the battery can be approximately determined by the electrolyte density.

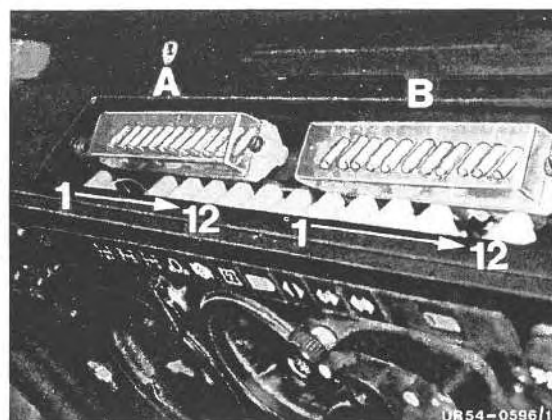
Electrolyte density in accordance with Baumé*	Specific weight*	Charged status of battery
32° Bé	1,285	well charged
24° Bé	1,20	half charged
16° Bé	1,12	discharged, charge immediately

* at an electrolyte temperature of +20°C

Electrolyte density (Tropics)

20°C		30°C		40°C		Charged status
Electrolyte density	Specific weight	Electrolyte density	Specific weight	Electrolyte density	Specific weight	
27,9° Bé	1,24	26,9° Bé	1,23	26° Bé	1,22	well charged
23°– 21° Bé	1,19 1,17	22°– 20° Bé	1,18 1,16	21°– 19° Bé	1,17– 1,15	half charged
14,2°– 10,6° Bé	1,11– 1,08	13°– 9,40° Bé	1,10– 1,07	11,2°– 8,0° Bé	1,09– 1,06	discharged charge immediately

Fuses



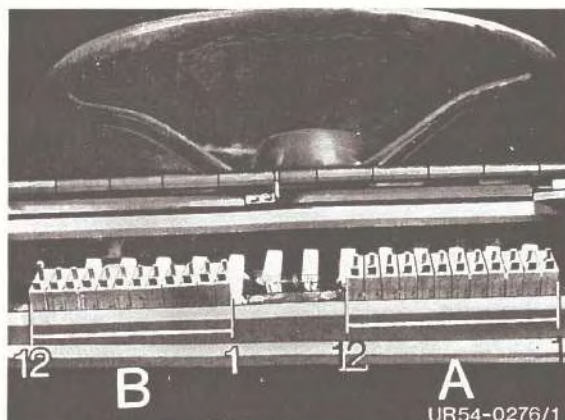
Fuse boxes A and B

Fuse No.	Fuse box A	Current Ampere
1	High beam, left/high beam indicator lamp	8
2	High beam, right	8
3	Low beam, left	8
4	Low beam, right/rear fog lamp	8
5	Shift gate indicator/stop light	8
6	Horn/switch exhaust brake	8
7	Turn signal right and left/hazard warning flasher switch	8
8	Windscreen wiper and washer unit	8
9	Heater/blower	16
10	Combination instrument revolution counter/radio connection	8
11	Indicator lamps/reversing lamp	8
12	–	–
Fuse No.	Fuse box B	Current Ampere
1	Parking light, left/switch illumination/marker lamp ¹⁾	8
2	Parking light, right/marker lamp ¹⁾	8
3	Tail light, left	8
4	Tail light, right/instrument illumination	8
5	Tachograph	8
6	Socket, driver's cab	8
7	Interior lamp/door contact	8
8	–	–
9	–	–
10	Hazard warning system	16
11	–	–
12	–	–
Overcurrent protective switch ¹⁾		Current Ampere
–	Power supply body	60

1) For SA 35 769/15/16

54.7 General

SA 35 769/10, SA 35 769/14, SA 35 769/17



Automatic cutouts A and B

Fuse No.	Automatic cutout A	Current Ampere
1	High beam, left /high beam indicator lamp	8
2	High beam, right	8
3	Low beam, left	8
4	Low beam, right (rear fog lamp ¹⁾)	8
5	Stop light	8
6	Signal horn	8
7	Windscreen washer	8
8	Windscreen wiper	8
9	Heater/blower	12
10	Combination instrument revolution counter/indicator lamps (reversing buzzer ²⁾)	8
11	Relay masked circuit/interior lamp driver's cab /shift gate indicator (Reversing lamp ¹⁾)	8
12	Hazard warning flasher	12

1) Applies only to SA 35 769/10 and SA 35 769/17 2) Applies only to SA 35 769/14

Fuse No.	Automatic cutout B	Current Ampere
1	Parking light, left/switch illumination - windscreen washer (Switch illumination - rear fog light ¹⁾)	8
2	Parking light, right/relay - shift gate indicator	8
3	Tail light, left (license plate light ¹⁾)	8
4	Tail light, right/instrument illumination (license plate light ²⁾)	8
5	Speedometer/map light	8
6	Socket, driver's cab	8
7	Socket, trailer 12-pole (socket rear 2-pole ¹⁾)	8
8	—	8
9	Convoy cross light	8
10	Masked tail light	8
11	Masked headlight, left	8
12	Masked headlight, right	8

1) Applies only to SA 35 769/10 and SA 35 769/14 2) Applies only to SA 35 769/17

Bulbs in accordance with DIN 72601

Power load	Output at 12 V W	at 24 V W	Form in accordance with DIN 72601	Remarks
Main headlamp	45/40	55/50	A	
Auxiliary headlamp for front attachments	45/40	55/50	A	
Halogen headlamp (right-hand traffic on)	60/55	75/70	H4	
Masked headlight	—	18	R	
Parking light	4	4	HL	
Side-marker lamp	5	5	L	
Turn signal, front	21	21	RL	
Auxiliary turn signal lights, front	4	4	HL	
Turn signal, rear	21	21	P 25 - 1	
Tail light	10	10	G	
Masked tail light	—	5	G	
Stop light	21	21	P 25 - 1	
Masked stop lamp	—	2	H	
License plate light	5	5	L	
License plate light (for masked brake circuit)	—	5	G	
Driver's cab interior lamp	10	10	K	
Rotating beacon ¹⁾	45	45	U	
Rotating beacon ²⁾ (Halogen)	55	70	H1	
Reversing lamp	35	35	D	
Convoy cross light		2	H	
Shift gate indicator lamp	1,2	1,2	W 5/1,2	
Clearance lamps	5	5	L	
Indicator lamps	2	2	H	
Combination instrument, hazard warning switch, speedometer, symbol indicator, rotating beacon, windscreen heated, battery charge indicator lamp				
Brake pad wear	—	2	H	
Indicator lamps, auxiliary headlamp for front attachments	4	4	HL	

1) To chassis end No. 087 827

2) From chassis end No. 087 828

Electrical Circuit Diagram (to chassis End No. 001 067)

Key to 1.5/2

1a	Turn signal, left	31	Fuse box B
1b	Turn signal, right	32	Fuse box A
2a	Main headlamp, left	33	Stop light switch
2b	Main headlamp, right	34	Control switch
3	Horn	35	Warning flasher
4	Light switch	36	Plug-and-socket connection (7 connectors each 8-pole)
5	Revolution counter	37	Switch for shift gate indicator
6	Combination instrument	38	Switch for parking brake
7	Tachograph	39	Switch for four-wheel drive
8	Ignition/starter switch	40	Switch for differential lock
9	Socket	41	Switch for exhaust brake
10	Regulating switch instrument illumination	42	Brake fluid reservoir A
11	Switch for windscreen washer	43	Brake fluid reservoir B
12	Driver's cab interior lamp	44	Sending unit for fuel supply
13	Switch cluster	45	Sending unit for oil pressure
14	Pump for windscreen washer	46	Sending unit for temperature (coolant)
15	Heater fan	47	Alternator
16	Wiper motor	48	Battery
17	Control switch for heater	49	Starter
18	Relay for starter	50a	Turn signal, tail light, stop lamp, left
19	Relay	50b	Turn signal, tail light, stop lamp, right
20	Shift gate indicator	51	Trailer socket (7 pole)
21	Indicator lamp for differential lock		
22	Indicator lamp for four - wheel drive		
23	Indicator lamp for parking brake		
24	Indicator lamp for brake fluid		
25	Indicator lamp for charging indicator		
26	Indicator lamp for high beam		
27	Turn signal indicator lamp for tractor		
28	Turn signal indicator lamp for 1st trailer		
29	Turn signal indicator lamp for 2nd trailer		
30	Hazard warning flasher switch		

m = Ground engine
 X = Ground
 Instrument panel
 y = Ground frame
 z = Ground driver's cab

Line Colour Codes:

bl = blue
 br = brown
 ge = yellow
 gn = green
 gr = grey
 li = violet
 rt = red
 sw = black
 ws = white



Electrical circuit diagram (from chassis end No. 001 068)

Key to 1.5/4

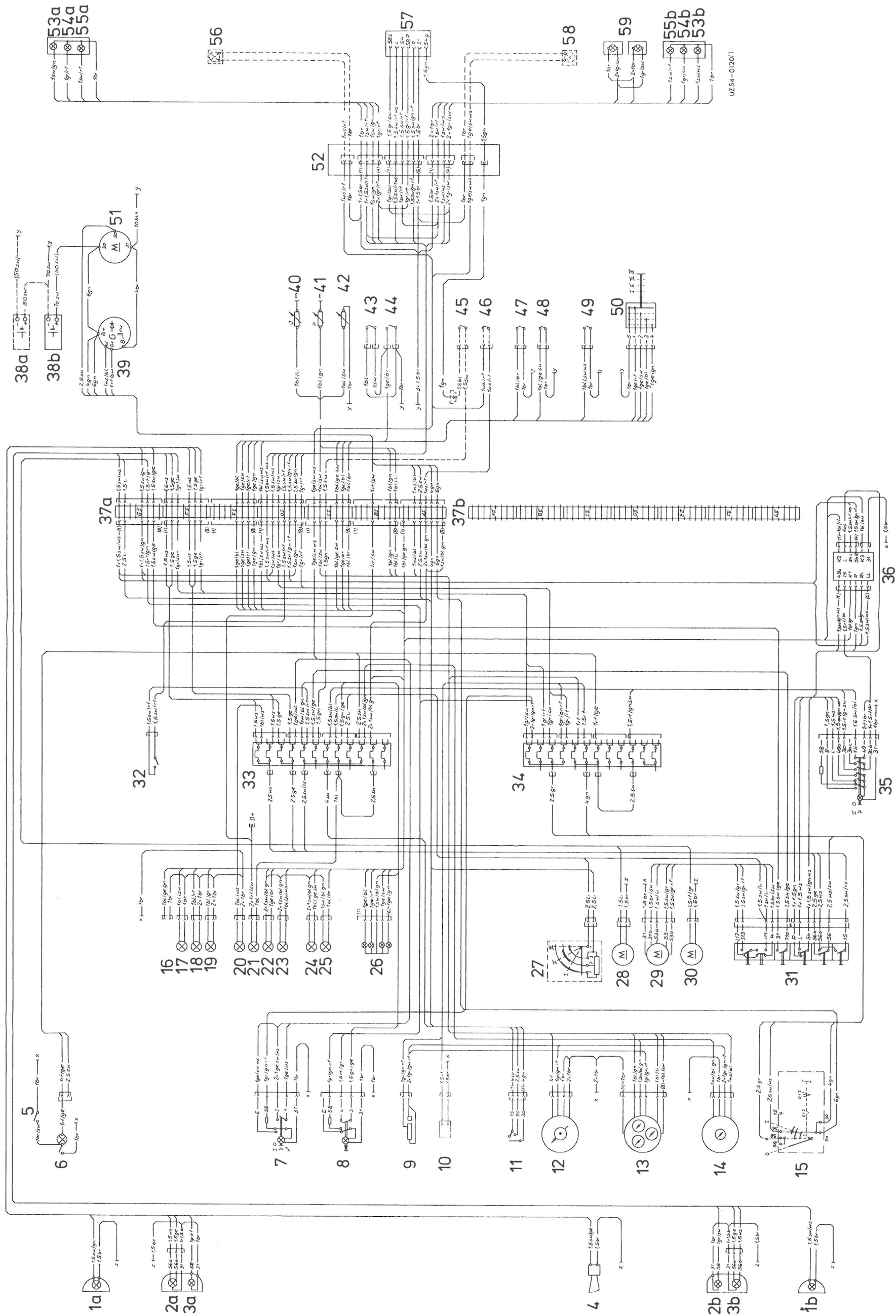
- | | | | |
|-----|---|-----|---------------------------------------|
| 1a | Direction indicator right | 38a | Battery (12 V system) |
| 1b | Direction indicator left | | Connection line 70 sw for 12 V |
| 2a | High beam/low beam right | | Connection line 50 sw for 24 V |
| 2b | High beam/low beam left | 38b | Battery (additional with 24 V system) |
| 3a | Parking light right | 39 | Generator |
| 3b | Parking light left | 40 | Pickup for temperature (coolant) |
| 4 | Signal horn | 41 | Pickup for oil pressure |
| 5 | Switch door contact | 42 | Pickup for fuel supply |
| 6 | Cab interior light | 43 | Brake fluid reservoir B |
| 7 | Switch, rear fog light | 44 | Brake fluid reservoir A |
| 8 | Switch for screen washer | 45 | Switch for engine brake |
| 9 | Rheostat for instrument panel lights | 46 | Switch, reversing light |
| 10 | Socket | 47 | Switch for differential lock |
| 11 | Ignition switch | 48 | Switch for 4-wheel drive |
| 12 | Tachograph | 49 | Switch for parking brake |
| 13 | Instrument cluster | 50 | Switch for shift indicator |
| 14 | Rev counter | 51 | Starter |
| 15 | Main light switch | 52 | Cable junction box rear |
| 16 | Connection for warning light, pto | 53a | Direction indicator right |
| 17 | Indicator warning light for second trailer | 53b | Direction indicator left |
| 18 | Indicator warning light for first trailer | 54a | Tail light right |
| 19 | Indicator warning light for tractor unit | 54b | Tail light left |
| 20 | Warning light for high beam | 55a | Brake light right |
| 21 | Warning light for charge indicator | 55b | Brake light left |
| 22 | Warning light for brake fluid | 56 | Reversing light ¹⁾ |
| 23 | Warning light for parking brake | 57 | Trailer socket (7-pin) |
| 24 | Warning light for 4-wheel drive | 58 | Rear fog light ¹⁾ |
| 25 | Warning light for differential lock | 59 | Licence plate light |
| 26 | Shift indicator | | |
| 27 | Control switch for heating | m | = Ground connection engine |
| 28 | Heater fan | X | = Ground connection instrument panel |
| 29 | Wiper motor | Y | = Ground connection frame |
| 30 | Pump for screen washer | Z | = Ground connection cab |
| 31 | Combination switch (wipers, signal horn, indicators, high/low beam, headlamp flasher) | | |
| 32 | Brake light switch | | |
| 33 | Fuse box A | | |
| 34 | Fuse box B | | |
| 35 | Warning flasher switch | | |
| 36 | Warning flasher pickup | | |
| 37a | Isolating point, cab | | |
| 37b | Isolating point, cab (not connected) | | |

Line colour code:

- bl = blue
 blk = blank
 br = brown
 ge = yellow
 gn = green
 gr = grey
 li = purple
 rs = pink
 rt = red
 sw = black
 ws = white

- 1) Retrofitting possible
 Special version

Electrical circuit diagram from engine end No. 001 068



Electrical Circuit Diagram (to chassis end No. 001 517)

Key to 1.5/6

1a	Turn signal, right	47b	Parting point driver's cab (not connected)
1b	Turn signal, left	48	Alternator
2a	High beam/ low beam, right	49	Sending unit coolant temperature
2b	High beam/low beam, left	50	Sending unit oil pressure gauge
3a	Parking light, right	51	Sending unit fuel gauge
3b	Parking light, left	52	Brake fluid reservoir II
4a	Masked headlight, right	53	Brake fluid reservoir I
4b	Masked headlight, left	54	Sending unit rear lamp
6	Horn	55	Sending unit differential lock
7	Switch door contact	56	Sending unit four-wheel drive
8	Interior lamp driver's cab	57	Sending unit parking brake
9	Sending unit for brake pad wear indicator front axle, left ¹⁾	58	Sending unit shift gate indicator
10	Switch, rear fog light	59	Starter
11	Switch, windscreen washer	60	Batteries
12	Regulating switch instrument illumination	61	Sending unit brake pad wear indicator, rear axle, left ¹⁾
13	Instrument panel 2-pole socket	62	Distributor box in frame side member, rear
14	Ignition switch	63a	Turn signal, right
15	Speedometer	63b	Turn signal, left
16	Revolution counter	64a	Tail light, right
17	Instrument cluster	64b	Tail light, left
18	Masked light switch	65a	Stop light, right
19	Reading light	65b	Stop light, left
20	Indicator lamp, PTO shaft	66a	Masked tail light, right
21	Indicator lamp, turn signal trailer	66b	Masked tail light, left
22	Indicator lamp, turn signal tractor	67	Convoy cross light
23	Indicator lamp, high beam	68	Socket 7-pole ¹⁾
24	Indicator lamp, charge indicator lamp	69	Socket 12-pole
25	Indicator lamp, brake fluid	70	Socket 2-pole
26	Indicator lamp, parking brake	71	Reversing lamp ¹⁾
27	Indicator lamp, four-wheel drive	72	Rear fog light ¹⁾
28	Indicator lamp, differential lock Front axle/rear axle	73	Masked stop lamp
29	Indicator lamp, brake pad wear indicator Rear axle ¹⁾	74a	License plate light, right
30	Indicator lamp, brake pad wear indicator Front axle ¹⁾	74b	License plate light, left
31	Indicator lamp, supply pressure		
32	Indicator lamp, shift gate indicator		
33	Diode group		
34	Control switch heater fan		
35	Heater fan - motor		
36	Wiper motor		
37	Pump windscreen washer		
38	Switch cluster (wiper, horn, turn signal, high, low beam, headlight flasher)		
39	Regulator		
40	Sending unit brake light		
41	Automatic cutout A		
42	Automatic cutout B		
43	Relay shift gate indicator bright - dark		
44	Relay masked light		
45	Hazard warning flasher switch		
46	Sending unit hazard warning system		
47a	Parting point driver's cab		

m	= Ground engine
X	= Ground Instrument panel
y	= Ground frame
z	= Ground driver's cab

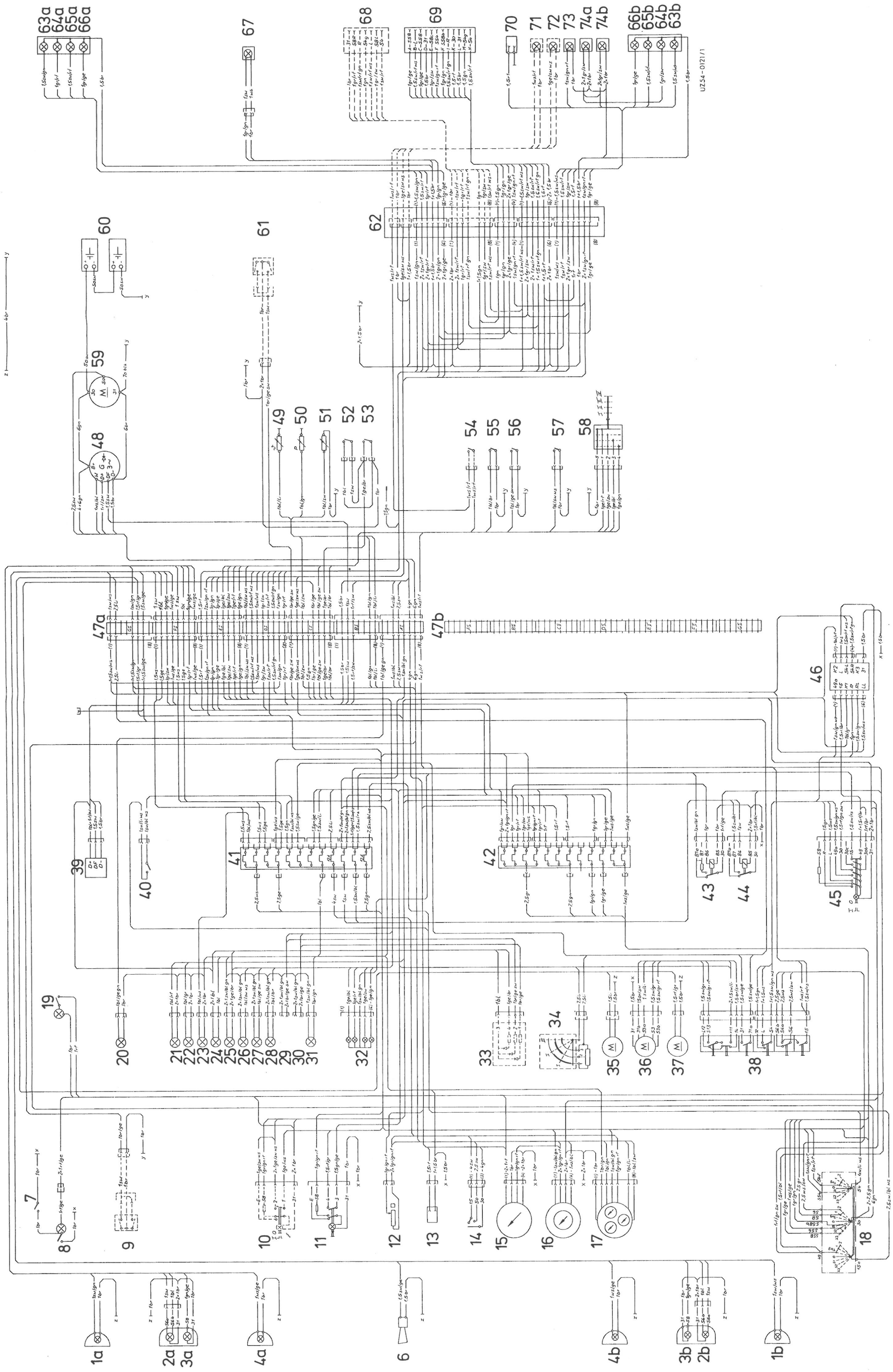
Line Colour Code:

bl	= blue
blk	= bright
br	= brown
ge	= yellow
gn	= green
gr	= grey
li	= violet
rs	= pink
rt	= red
sw	= black
ws	= white

1) Subsequent installation possible

--- Special version

Electrical circuit diagram (up to chassis end No. 001 517)



Electrical circuit diagram (from engine end No. 001 518)

Key to 1.5/8

1a	Direction indicator right	46	Generator
1b	Direction indicator left	47	Switch, differential pressure ¹⁾
2a	High beam/low beam right	48	Pickup, coolant temperature
2b	High beam/low beam left	49	Pickup, oil pressure gauge
3a	Parking light right	50	Pickup, fuel gauge
3b	Parking light left	51	Brake fluid reservoir II
4a	Masked headlamp right	52	Brake fluid reservoir I
4b	Masked headlamp left	53	Pickup, reversing light
5	Signal horn	54	Pickup, differential lock
6	Switch, door contact	55	Pickup, 4-wheel drive
7	Cab interior light	56	Pickup, parking brake
8	Pickup for brake lining wear indicator front axle left ¹⁾	57	Pickup, shift indicator
9	Switch, rear fog light	58	Starter
10	Switch, screen washer	59	Batteries
11	Sheostat instrument panel lights	60	Pickup, brake lining wear indicator rear axle left ¹⁾
12	Socket 2-pin, instrument panel	61	Junction box in side member at rear
13	Ignition switch	62a	Direction indicator right
14	Speedometer	62b	Direction indicator left
15	Rev counter	63a	Tail light right
16	Instrument cluster	63b	Tail light left
17	Masked light switch	64a	Brake light right
18	Reading light	64b	Brake light left
19	Warning light, pto	65a	Masked tail light right
20	Warning light, trailer direction indicators	65b	Masked tail light left
21	Warning light, tractor unit direction indicators	66	Convoy cross light
22	Warning light, high beam	67	Socket 7-pin ¹⁾
23	Warning light, charge indicator	68	Socket 12-pin
24	Warning light, brake fluid or differential pressure	69	Socket 2-pin
25	Warning light, parking brake	70	Reversing light ¹⁾
26	Warning light, 4-wheel drive	71	Rear fog light ¹⁾
27	Warning light, differential lock front axle/rear axle	72	Masked brake light
28	Warning light, brake lining wear indicator rear axle ¹⁾	73a	Licence plate light right
29	Warning light, brake lining wear indicator front axle ¹⁾	73b	Licence plate light left
30	Warning light, supply pressure		
31	Warning light, shift indicator	m	= Ground connection engine
32	Diode group	X	= Ground connection instrument panel
33	Control switch, heater fan	Y	= Ground connection frame
34	Heater fan motor	Z	= Ground connection cab
35	Wiper motor		
36	Pump, screen washer		
37	Combination switch (wipers, signal horn, indicators, high/low beam, headlamp flasher)		
38	Pickup, brake lights		
39	Automatic cutout A		
40	Automatic cutout B		
41	Relay, shift indicator light/dark		
42	Relay, masked light		
43	Hazard warning switch		
44	Pickup, hazard warning system		
45a	Isolating point, cab		
45b	Isolating point, cab (not connected)		

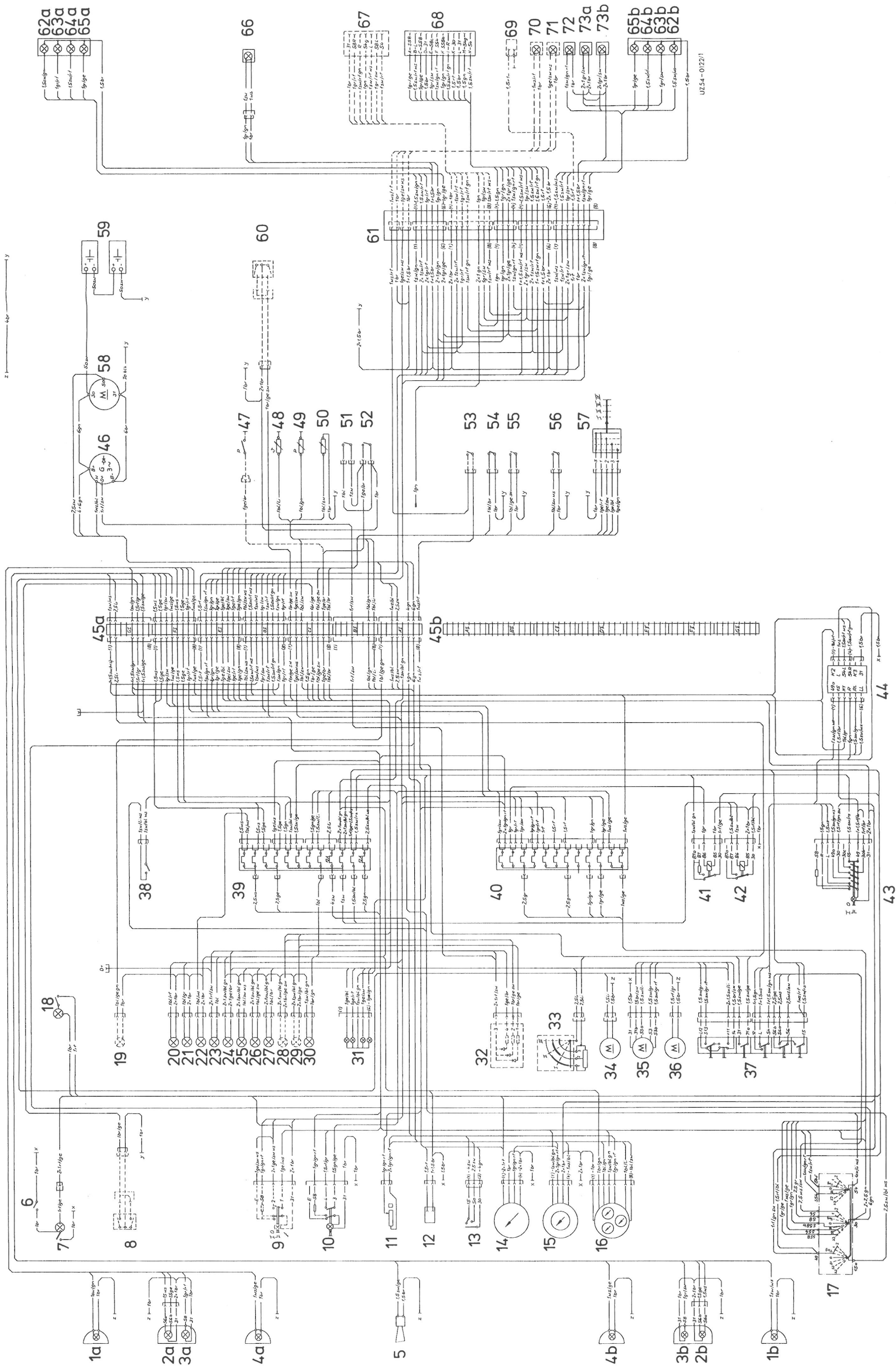
m = Ground connection engine
 X = Ground connection instrument panel
 Y = Ground connection frame
 Z = Ground connection cab

Line colour code:

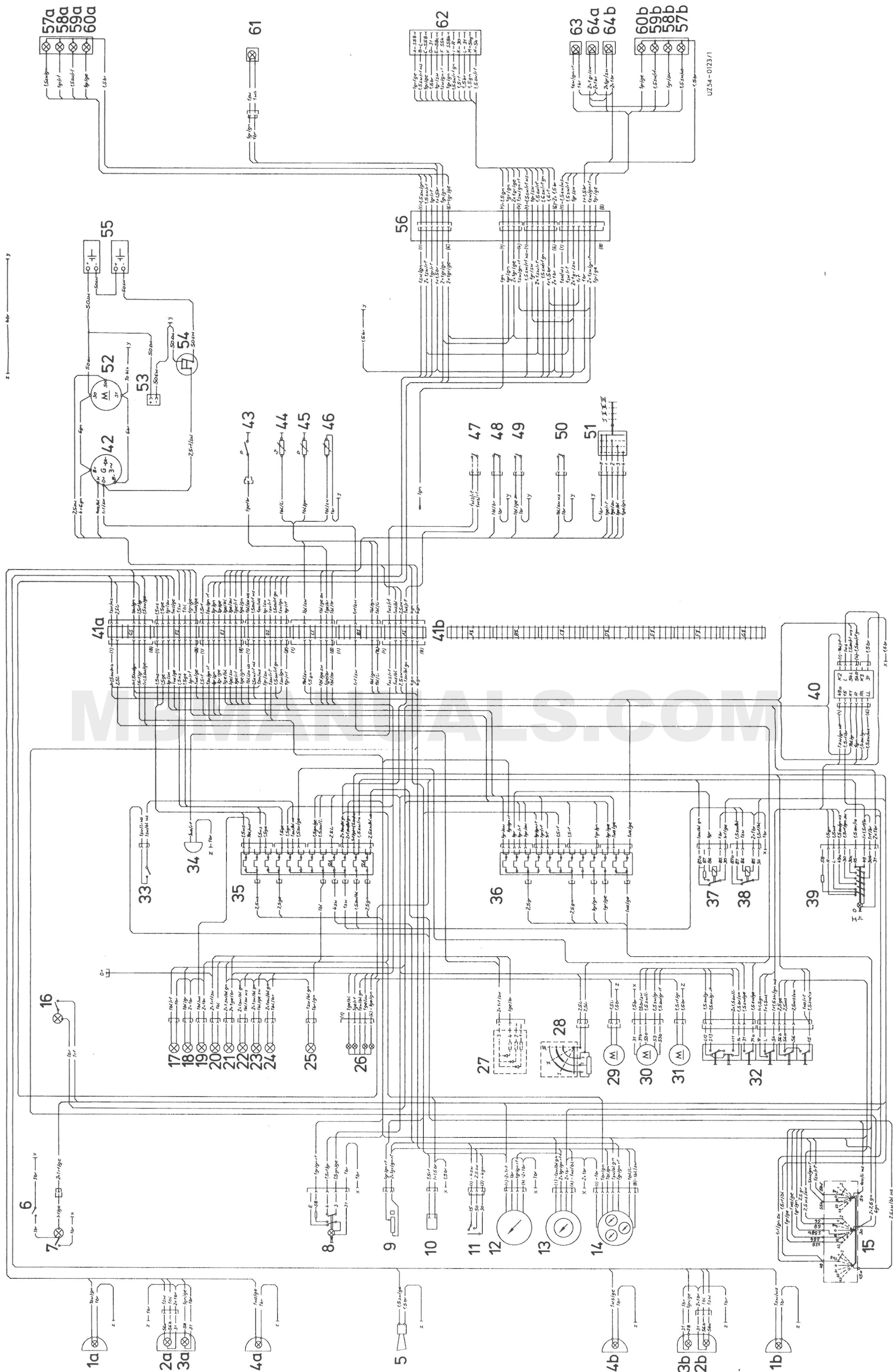
bl = blue
 blk = blank
 br = brown
 ge = yellow
 gn = green
 gr = grey
 li = purple
 rs = pink
 rt = red
 sw = black
 ws = white

1) Retrofitting possible
 Special version

Electrical circuit diagram (from chassis end No. 001 518)



Electrical circuit diagram



Key to 1.5/12

- | | | | |
|----|---|-----|---------------------------------------|
| 1a | Direction indicator right | 37 | Brake light switch |
| 1b | Direction indicator left | 38 | Fuse box A |
| 2a | High beam/low beam right | 39 | Fuse box B |
| 2b | High beam/low beam left | 40 | Warning flasher switch |
| 3a | Parking light right | 41 | Warning flasher pickup |
| 3b | Parking light left | 42a | Isolating point, cab |
| 4a | Marker lights right | 42b | Isolating point, cab (not connected) |
| 4b | Marker lights left | 43 | Battery (2 x 12 V) |
| 5 | Switch, door contact | 44 | Generator |
| 6 | Cab interior light | 45 | Socket, power supply (body) |
| 7 | Signal horn | 46 | Pickup for temperatur (coolant) |
| 8 | Overcurrent cutout switch (power supply) | 47 | Pickup for oil pressure |
| 9 | Switch, rear fog light | 48 | Pickup for fuel supply |
| 10 | Switch, screen washer | 49 | Switch, differential pressure |
| 11 | Rheostat for instrument panel lights | 50 | Switch for engine brake ¹⁾ |
| 12 | Map reading light | 51 | Switch, reversing light ¹⁾ |
| 13 | Socket | 52 | Switch for differential lock |
| 14 | Ignition switch | 53 | Switch for 4-wheel drive |
| 15 | Tachograph | 54 | Switch for parking brake |
| 16 | Instrument cluster | 55 | Switch for shift indicator |
| 17 | Rev counter, electronic | 56 | Starter |
| 18 | Main light switch | 57 | Cable junction box rear |
| 19 | Connection for warning light, pto | 58a | Indicator light right |
| 20 | Indicator warning light for second trailer | 58b | Indicator light left |
| 21 | Indicator warning light for first trailer | 59a | Tail light right |
| 22 | Indicator warning light for tractor unit | 59b | Tail light left |
| 23 | Warning light for high beam | 60a | Brake light right |
| 24 | Warning light for charging indicator | 60b | Brake light left |
| 25 | Warning light for differential pressure | 61 | Licence plate lights |
| 26 | Warning lifht for parking brake | 62 | Rear fog light ¹⁾ |
| 27 | Warning light for 4-wheel drive | 63 | Trailer socket (7-pin) |
| 28 | Warning light for differential lock | 64 | Reversing light ¹⁾ |
| 29 | Shift indicator | | |
| 30 | Isolating relay, power supply | m | = Ground connection engine |
| 31 | Relay, power supply | X | = Ground connection instrument panel |
| 32 | Switch, heater fan | Y | = Ground connection frame |
| 33 | Heater fan | Z | = Ground connection cab |
| 34 | Wiper motor | | |
| 35 | Pump for screen washer | | |
| 36 | Combination switch (wipers, signal horn, indicators, high/low beam, headlamp flasher) | | |
| | | | Line colour code: |
| | | bl | = blue |
| | | blk | = blank |

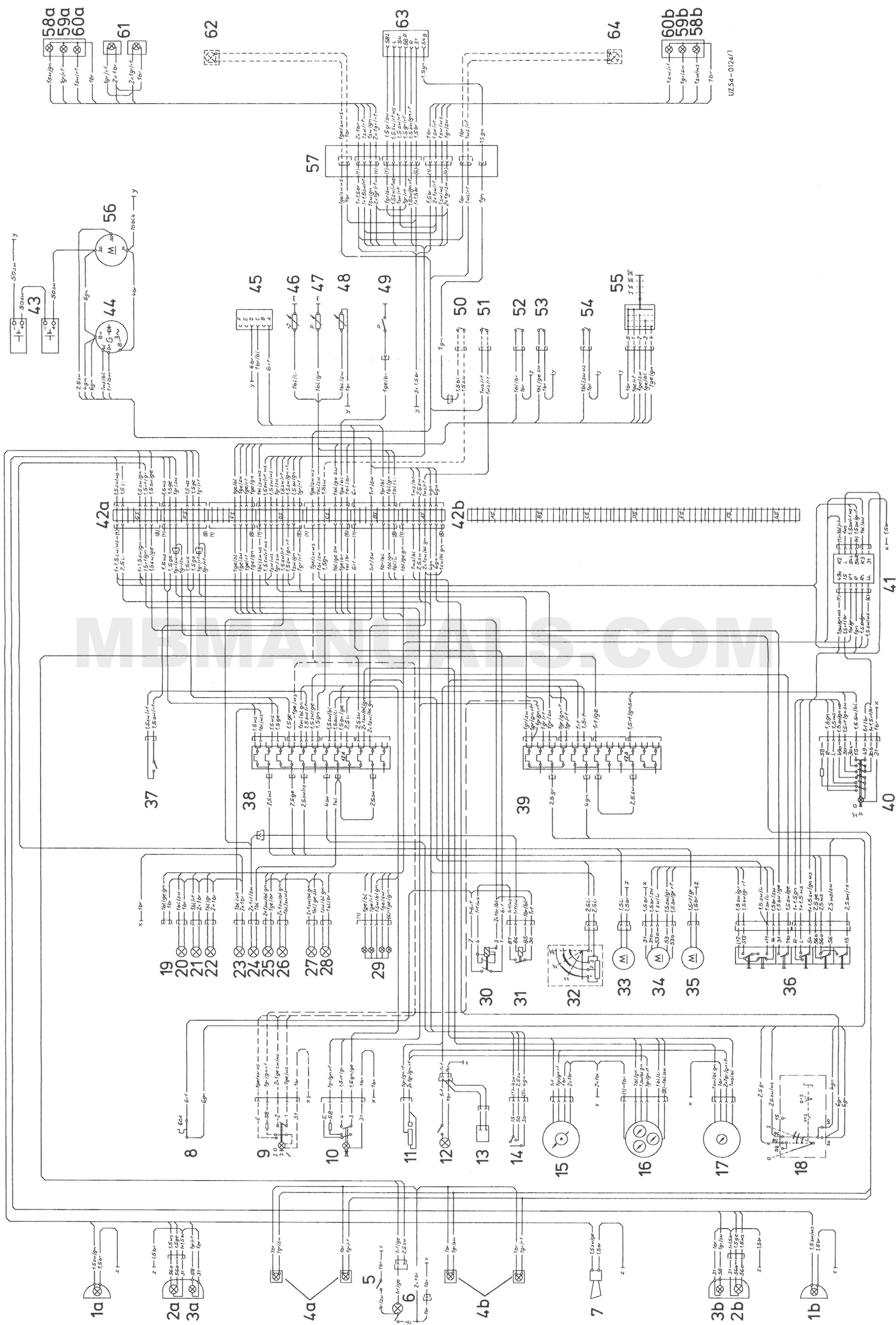
m = Ground connection engine
X = Ground connection instrument panel
Y = Ground connection frame
Z = Ground connection cab

Line colour code:

bl	=	blue
blk	=	blank
br	=	brown
ge	=	yellow
gn	=	green
gr	=	grey
li	=	purple
rs	=	pink
rt	=	red
sw	=	black
ws	=	white

- 1) Retrofitting possible
- Special version

Electrical circuit diagram



Electrical circuit diagram

Key to 1.5/14

1a	Direction indicator right	45a	Isolating point, cab
1b	Direction indicator left	45b	Isolating point, cab (not connected)
2a	High beam/low beam right	46	Alternator
2b	High beam/low beam left	47	Speed pulse pickup
3a	Parking light right	48	Switch, differential pressure
3b	Parking light left	49	Pickup, coolant temperature
4a	Masked headlight right	50	Pickup, oil pressure gauge
4b	Masked headlight left	51	Pickup, fuel gauge
5	Signal horn	52	Switch, reversing light
6	Switch, door contact	53	Pickup, differential lock
7	Interior light, cab	54	Pickup, four-wheel drive
8	Pickup for brake lining wear indicator, front axle left	55	Pickup, parking brake
9	Switch, rear fog light	56	Pickup, shift indicator
10	Switch, windshield washer	57	Starter
11	Dimmer, instrument lights	58	Batteries
12	Plug socket 2-pole, instrument panel	59	Pickup for brake lining wear indicator, rear axle left
13	Ignition switch	60	Junction box in side member rear
14	Speedometer	61a	Direction indicator right
15	Rev. counter	61b	Direction indicator left
16	Instrument cluster	62a	Tail light right
17	Masked light switch	62b	Tail light left
18	Reading lamp	63a	Stop light right
19	Indicator lamp, pto shaft ¹⁾	63b	Stop light left
20	Indicator lamp, direction indicator 1st trailer	64a	Masked tail light right
21	Indicator lamp, direction indicator tractor unit	64b	Masked tail light left
22	Indicator lamp, high beam	65a	Licence plate light right
23	Connection for charging indicator lamp	65b	Licence plate light left
24	Indicator lamp, differential pressure	66	Masked stop light
25	Indicator lamp, parking brake	67	Convoy cross light
26	Indicator lamp, four-wheel drive	68	Reversing light
27	Indicator lamp, differential lock front axle/rear axle	69	Rear fog light
28	Indicator lamp, brake pad wear indicator, rear axle left	70	Plug socket 7-pole
29	Indicator lamp, brake pad wear indicator, front axle left	71	Plug socket 12-pole
30	Indicator lamp, supply pressure		
31	Indicator lamp, shift indicator	m	= Ground connection engine
32	Diode group	X	= Ground connection instrument panel
33	Control switch for blower	Y	= Ground connection frame
34	Heater blower motor	Z	= Ground connection cab
35	Wiper motor		
36	Pump, windshield washer		
37	Combination switch (wipers, horn, direction indicators, high/low beam, headlight flasher)		
38	Brake light switch		
39	Automatic fuse A		
40	Automatic fuse B		
41	Relay, shift indicator light/dark		
42	Relay, masked light		
43	Hazard warning switch		
44	Pickup, hazard warning system		

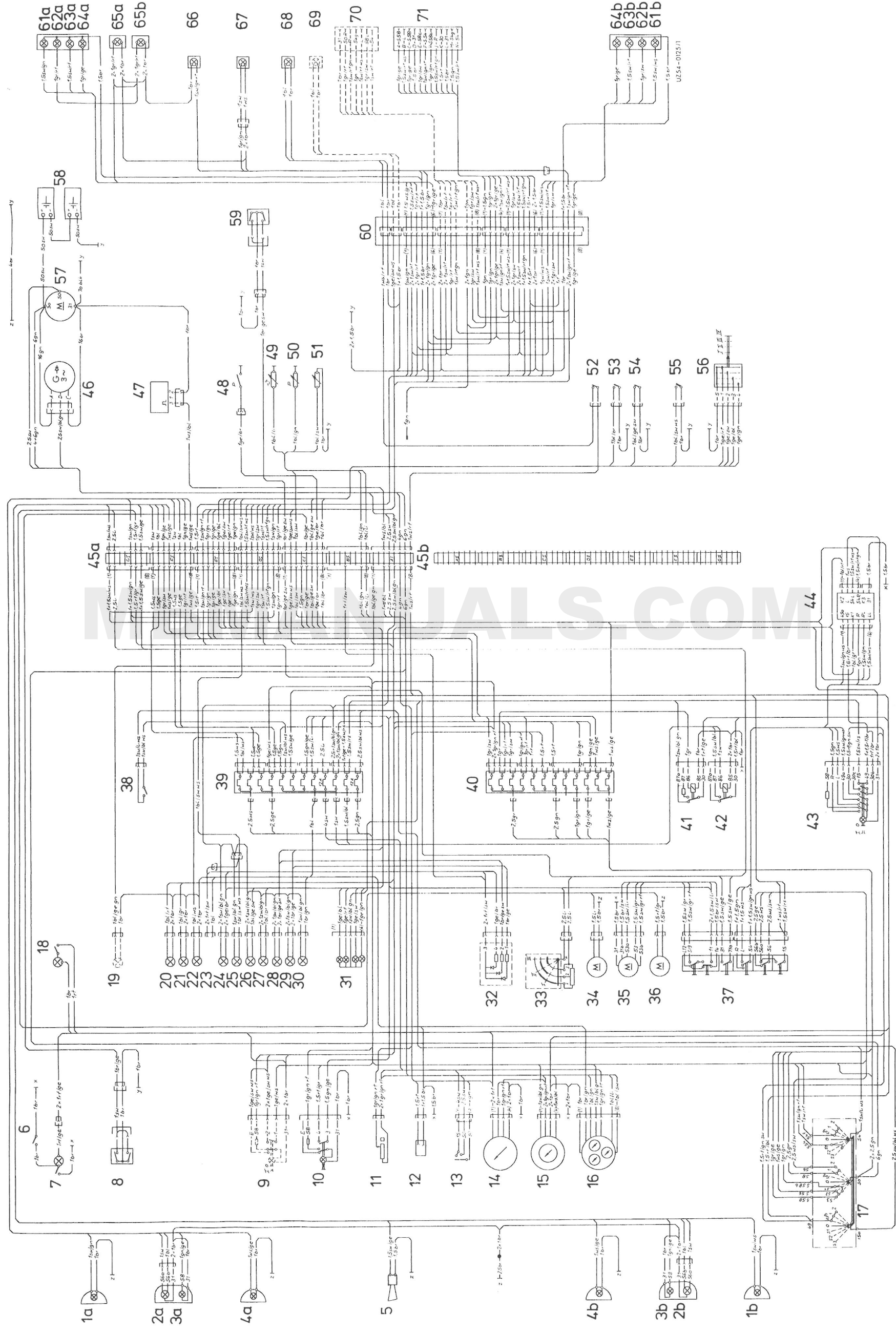
Line colour coding:

bl	= blue
blk	= blank
br	= brown
ge	= yellow
gn	= green
gr	= grey
li	= purple
rs	= pink
rt	= red
sw	= black
ws	= white

1) Retrofitting possible

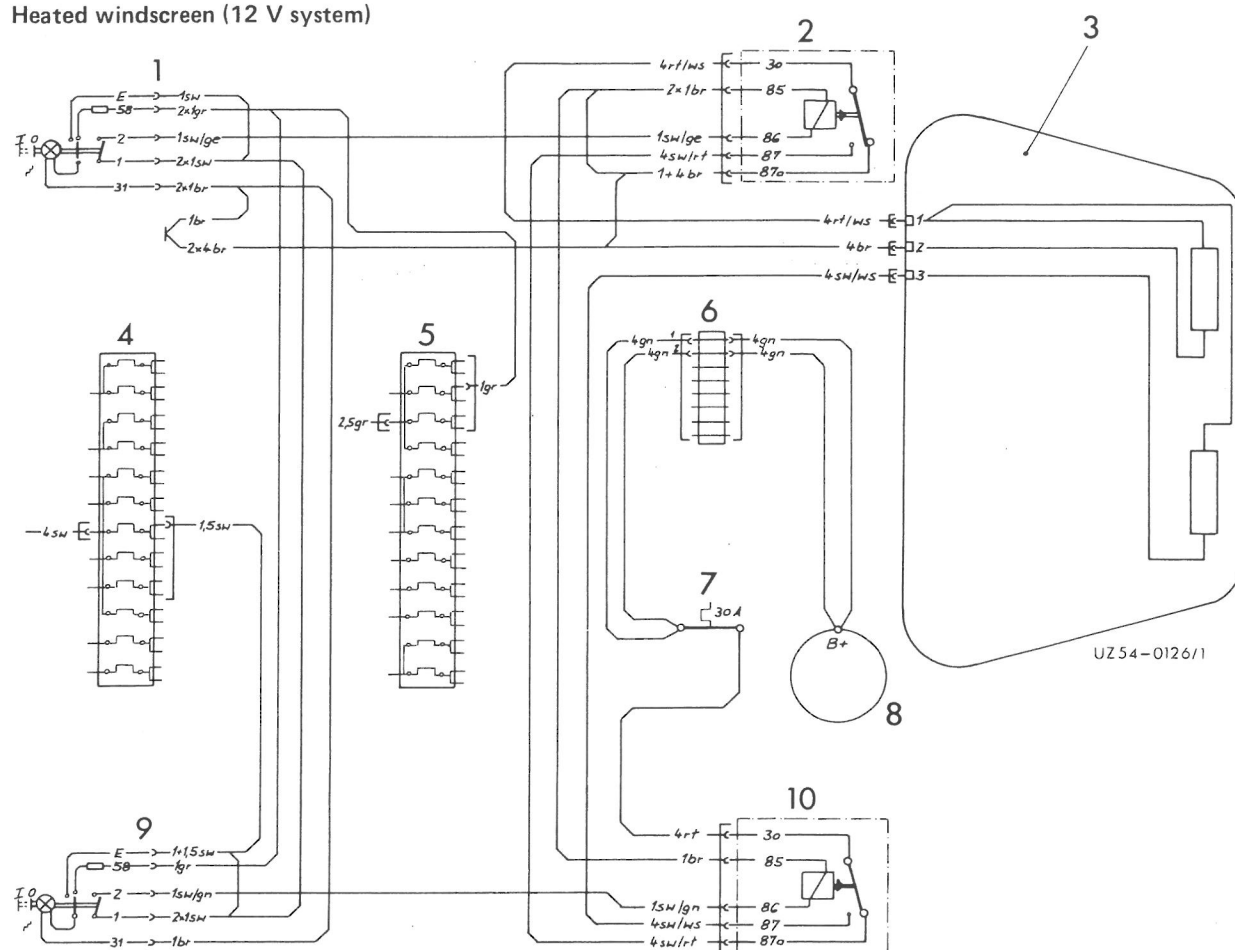
— — Optional extra

Electrical circuit diagram



Electrical circuit diagram

Heated windscreen (12 V system)



1 Switch) for heating panel (driver's side)

2 Relay)

3 Windscreen

Connections 1 and 2 driver's side

Connections 1 and 3 co-driver's side

Connections 2 and 3 full heating panel

4 Fuse box B (right)

5 Fuse box A (left)

6 Isolating point cab bulkhead

7 Overcurrent cutout 30 A

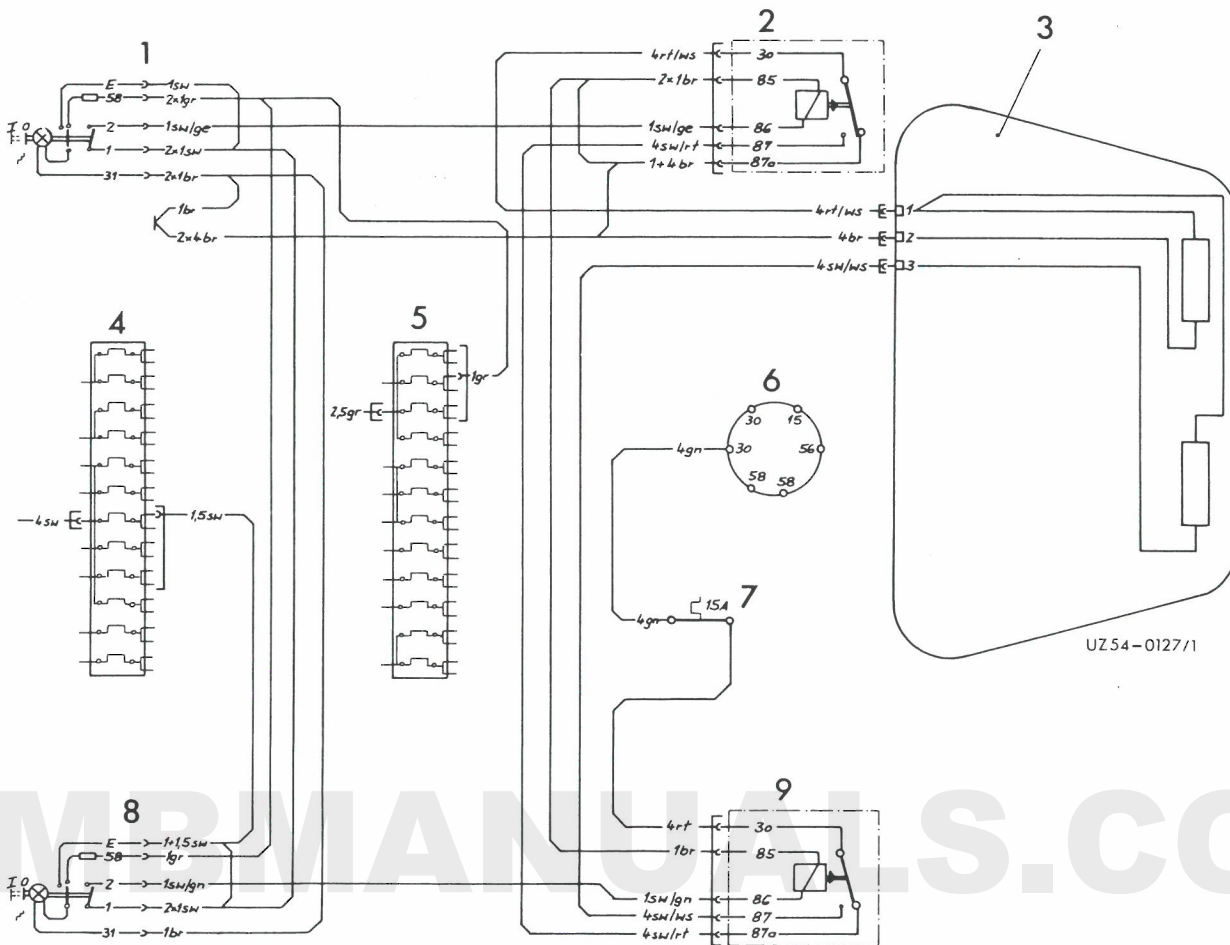
8 Generator

9 Switch) for heating panel (co-driver's side)

10 Relay)

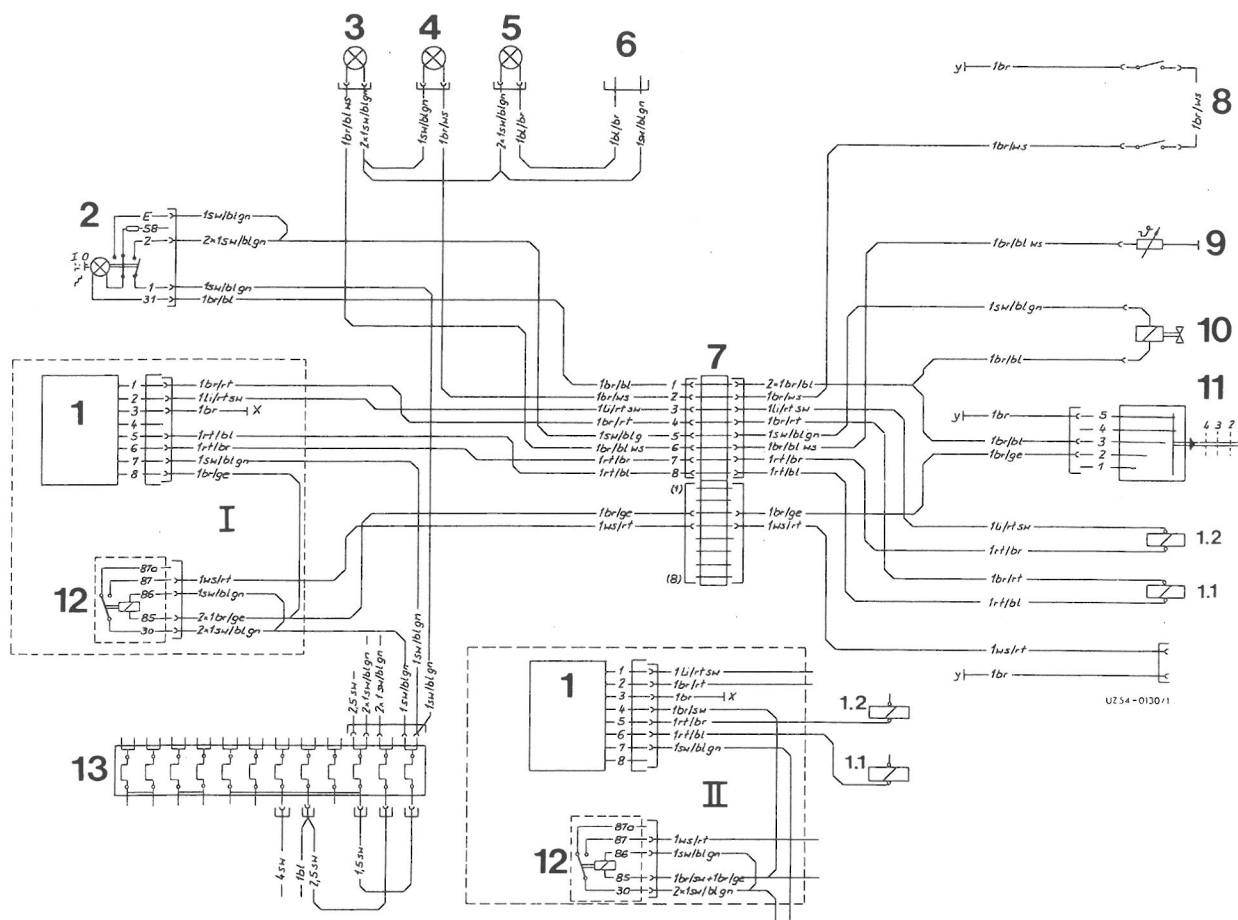
Electrical circuit diagram

Heated windscreen (24 V system)



- | | | | |
|---|---|---|--|
| 1 | Switch) | 6 | Ignition switch |
| 2 | Relay) for heating panel (driver's side) | 7 | Overcurrent cutout 15 A |
| 3 | Windscreen | 8 | Switch) |
| | Connections 1 and 2 driver's side | 9 | Relay) for heating panel (co-driver's side) |
| | Connections 1 and 3 co-driver's side | | |
| | Connections 2 and 3 full heating panel | | |
| 4 | Fuse box B (right) | | |
| 5 | Fuse box A (left) | | |

Electrical circuit diagram hydrostat

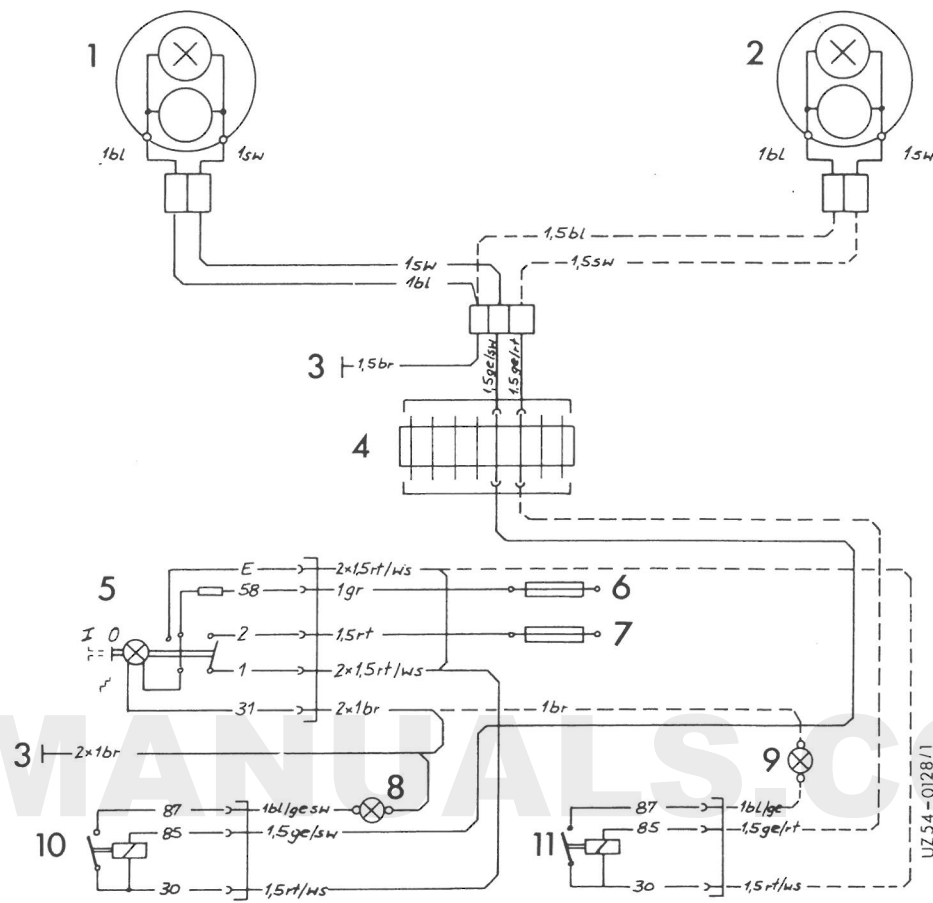


- I = Electrical connection at pickup with lefthand steering
 II = Electrical connection at pickup with righthand steering

- 1 Pickup, electronic
 1.1 Control unit, forward
 1.2 Control unit, reverse
 2 Switch, hydrostat
 3 Warning lamp, oil temperatur
 4 Switch-on warning lamp, hydrostat
 5 Warning lamp, differential lock
 6 Plug connection, differential lock

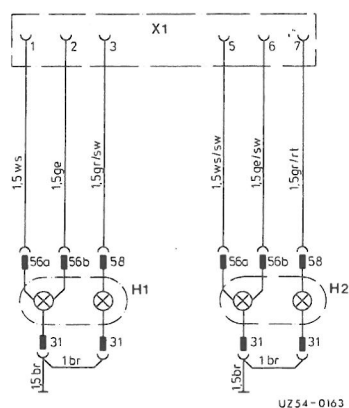
- 7 Isolating point (second test plate)
 8 Switch, cylinder
 9 Pickup, oil temperature
 10 Solenoid valve, pneumatic circuit
 11 Switch, interlock forward/reverse
 12 Relay, reversing lamp
 I = connection with lefthand steering
 II = connection with righthand steering
 13 Fuse box A
 (see 1.5/1 – 1.5/14)

Electrical circuit diagram rotating beacon



- | | | | |
|---|--|----|---------------------|
| 1 | Rotating beacon left | 7 | Fuse, termin 30 |
| 2 | Rotating beacon right | 8 | Warning lamp left |
| 3 | Ground connection bulkhead | 9 | Warning lamp right |
| 4 | Isolating point, second test plate on bulkhead | 10 | Control relay left |
| 5 | Switch | 11 | Control relay right |
| 6 | Fuse, terminal 58 | | |

Auxiliary Headlamp



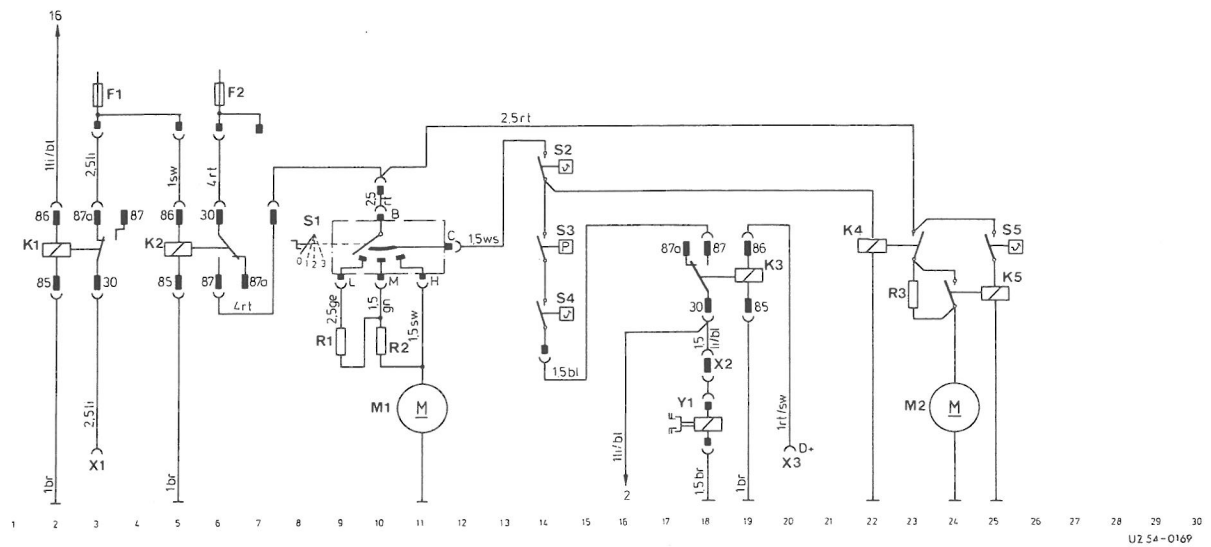
- H1 Auxiliary headlamp, right
H2 Auxiliary headlamp, left
X1 Plug-and-socket connection FI

Line Colour Code:

br = brown
ge = yellow
gr = grey
rt = red
sw = black
ws = white

Z = Ground driver's cab

Air Conditioning System 12 V



- F1 Fuse box A
- F2 Fuse box B
- K1 Relay heater off
- K2 Relay switch, air conditioning system
- K3 Relay D + Control
- K4 Relay
- K5 Relay
- M1 Evaporater
- M2 Condenser
- R1 Series resistor 1,8 Ohm

- | | |
|----|---|
| R2 | Series resistor 0.8 Ohm |
| R3 | Series resistor 0.5 Ohm |
| S1 | Switch, air conditioning system |
| S2 | Temperature switch |
| S3 | Pressure switch |
| S4 | Temperature switch |
| S5 | Temperature switch |
| X1 | to switch heater, ventilator |
| X2 | Plug-and-socket connection parting point driver's cab |
| X3 | Connection D + |
| Y1 | Solenoid coupling |

SA 35 914
SA 53 970

- | | | | |
|----|--------------------------------------|----|---|
| F1 | Fuse box A | R3 | Series resistor 1.2 Ohm |
| F2 | Fuse box B | R4 | Series resistor 3.4 Ohm |
| K1 | Relay heater off | S1 | Switch, air conditioning system |
| K2 | Relay switch air conditioning system | S2 | Temperature switch |
| K3 | Relay D + Control | S3 | Pressure switch |
| K4 | Relay | S4 | Temperature switch |
| K5 | Relay | S5 | Temperature switch |
| M1 | Evaporater | X1 | to switch heater, ventilator |
| M2 | Condenser | X2 | Plug-and-socket connection parting point driver's cab |
| R1 | Series resistor 4.8 Ohm | X3 | Connection D + |
| R2 | Series resistor 1.8 Ohm | Y1 | Solenoid coupling |